AI ASSISTANT CODING –ASS-9.2

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BATCH:39

TASK-1:

```python
# Write a short and technically accurate docstring for each function in this Python script.
# Keep the summaries concise.
# Do not explain implementation details.

# def calculate_average(numbers):
#     total = sum(numbers)
#     return total / len(numbers)

# def find_max(numbers):
#     return max(numbers)

# def filter_even(numbers):
#     return [num for num in numbers if num % 2 == 0]
def calculate_average(numbers):
    """Calculate and return the average of a list of numbers."""
    total = sum(numbers)
    return total / len(numbers)

def find_max(numbers):    """Return the maximum value from a list of numbers."""
    return max(numbers)
def filter_even(numbers):    """Return a list of even numbers from the given list."""
    return [num for num in numbers if num % 2 == 0]
```

TASK 2:

```python
# Add comments explaining only the logic behind the conditional statements and loops.
# Do not explain basic Python syntax.

# def categorize_numbers(numbers):
#     result = {"positive": [], "negative": [], "zero": []}

#     for num in numbers:
#         if num > 0:
#             result["positive"].append(num)
#         elif num < 0:
#             result["negative"].append(num)
#         else:
#             result["zero"].append(num)

#     return result

def categorize_numbers(numbers):
    """Categorize numbers into positive, negative, and zero."""
    result = {"positive": [], "negative": [], "zero": []}

    for num in numbers:
        # Check if the number is greater than zero and categorize as positive
        if num > 0:
            result["positive"].append(num)
        # Check if the number is less than zero and categorize as negative
        elif num < 0:
            result["negative"].append(num)
        # If the number is neither positive nor negative, categorize as zero
        else:
            result["zero"].append(num)

    return result
```

TASK 3 & 4:

```python
#  Write a concise file-level overview describing the purpose and functionality of this Python file.
# Place the overview as a module-level docstring at the top.
# Keep it brief and professional.
"""This Python file contains functions for performing basic operations on lists of numbers,
 including calculating the average, finding the maximum value, filtering even numbers, and categorizing numbers into positive,
 negative, and zero. Each function is designed to handle a list of numerical inputs and return the appropriate results based on the
 specified criteria. The functions are intended for use in various applications where numerical data processing is required, providing
   efficient and straightforward solutions for common tasks involving lists of numbers."""
# Rewrite the existing comments and docstrings to improve clarity, consistency, and professionalism.
# Do not change the technical meaning of the code.
# Keep the documentation concise and clear.
"""This module provides functions for performing basic operations on lists of numbers, including calculating the average,
finding the maximum value, filtering even numbers, and categorizing numbers into positive, negative, and zero. Each function is designed to handle a
list of numerical inputs and return the appropriate results based on the specified criteria. The functions are intended for use in various applications
where numerical data processing is required, providing efficient and straightforward solutions for common tasks involving lists of numbers."""
```

```python
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def calculate_average(numbers):    """Calculate and return the average of a list of numbers."""
    total = sum(numbers)
    return total / len(numbers)
def find_max(numbers):    """Return the maximum value from a list of numbers."""
    return max(numbers)
def filter_even(numbers):    """Return a list of even numbers from the given list."""
    return [num for num in numbers if num % 2 == 0]
def categorize_numbers(numbers):    """Categorize numbers into positive, negative, and zero."""
    result = {"positive": [], "negative": [], "zero": []}

    for num in numbers:
        # Check if the number is greater than zero and categorize as positive
        if num > 0:
            result["positive"].append(num)
        # Check if the number is less than zero and categorize as negative
        elif num < 0:
            result["negative"].append(num)
        # If the number is neither positive nor negative, categorize as zero
        else:
            result["zero"].append(num)

    return result
```

TASK 5:

```python
# Write a clear and technically accurate docstring for this function.
# Include:
# - The purpose of the function
# - Explanation of return value
# - Edge case handling
# Keep it concise but complete.
def calculate_average(numbers):
    """Calculate and return the average of a list of numbers.

    This function takes a list of numerical values and computes their average by summing the values and dividing by the count of numbers.
    If the input list is empty, it will raise a ZeroDivisionError.

    Args:
        numbers (list): A list of numerical values.
    Returns:        float: The average of the numbers in the list.
    Raises:        ZeroDivisionError: If the input list is empty.
    """
    total = sum(numbers)
    return total / len(numbers)
```